

Mobile-Based Infant Jaundice Screening Using Vision Transformer and MLP

Seema G S

Assistant Professor

Email: seemags@rnsit.ac.in

Amrutha M

Student

Email: amrutham23aiml@rnsit.ac.in

Divyashree

Student

Email: divyashree23aiml@rnsit.ac.in

Department of CSE (AI & ML), RNS Institute of Technology, Bangalore, India

Abstract—needs frequent diagnosis for preventing serious conditions such as kernicterus and hearing loss. Traditional diagnostic methods like blood tests and other devices are effective, invasive and less accessible for all resource settings. The project we worked on JaundiceNet: AI-Based Mobile Screening for Infant Jaundice, presents non-invasive, less cost and flexible solutions using artificial technology and mobile technology. This applications uses Vision Transformer (ViT) to extract visual features from facial and scleral images of infants, and also for classification through Multi-layer Perceptron (MLP). Lighting normalization, region cropping and image resizing are the initial steps. For clinical reliability the model is trained on neonatal datasets to get more accuracy. It is designed for real-time mobile implementation, JaundiceNet aims especially in rural and underserved areas. Future improvements consists of expanding datasets, improved processing for various lighting.

Index Terms—

I. INTRODUCTION

One of the most common medical conditions affecting infants is Neonatal jaundice, it occurring typically within first week of birth. Increased levels of bilirubin—a pigment released during breakdown of red blood cells causes yellowish tint in the skin and eyes. Mild jaundice can be cured naturally, but sever cases leads to brain damage, hearing loss and developmental delays if not treated early. In healthy babies, bilirubin is processed by the liver and discharged through stool. However, I infants the liver function is not completely developed, resulting in increased bilirubin level. In the first few days after birth this makes early detection critical when risk is high. Throughout the world, healthcare providers provide screening facility both in hospitals and homes, which also helps in educating the parents about signs of jaundice. Blood test which is traditional diagnostic method, which are accurate but they are high-risk and resource-intensive. Visual checks can be unreliable which may vary from one person to other depending on baby's skin tone and lighting condition.

Transcutaneous bilirubinometers offer a non-invasive alter native but are expensive and less accessible in low resource settings. To overcome these challenges, this project developed a user-friendly, AI-powered tool which uses smartphone cameras to capture facial and scleral images of infants. To detect jaundice with high accuracy these images are analysed by trained deep learning models. The system is developed to be non-invasive, cost-effective and deployable in both healthcare

centres and homes. By improving the power of AI with the convenience of mobile technology, this approach helps to reach more families making early detection of neonatal jaundice easier. It improves outcomes and lessens the strain on healthcare systems by encouraging early intervention, lowering reliance on specialized equipment, and providing caregivers with a trustworthy tool for tracking infant health.

II. LITERATURE REVIEW

[1] Looks into image-based alternatives for monitoring neonatal jaundice in order to overcome the drawbacks of invasive blood sampling and the high expense of standard equipment. The review covers everything from sophisticated AI applications like deep learning and risk prediction using the Bhutani nomogram to more conventional diagnostic techniques like visual evaluation and serum bilirubin measurement. The viability of portable handheld technology is one of the paper's main topics. It states that precise bilirubin estimation with common smartphones is made possible by combining image processing with technologies like Artificial Neural Networks (ANN) and Wireless Sensor Networks (WSN). Interestingly, one rule-based system that used these techniques achieved an accuracy rate of 91.7%.

Neonatal jaundice must be identified as soon as possible to prevent serious long-term effects like brain damage. Recent studies [2] tested techniques like SVR, KNN, and linear regression to see how AI can help. Deep learning models outperformed these conventional machine-learning methods in this comparison. Deep neural networks may be the best prediction tool, as demonstrated by the ResNet model's 96% accuracy rate. The study also found that, with a correlation of 0.91, Gaussian Process Regression (GPR) provided the best results. The study also looked at readily available screening resources like the BiliCam app for smartphones and other techniques that use sclera imaging or color calibration.

Researchers in [3] created a smartphone-integrated system that uses diffuse reflectance and neural networks to measure bilirubin levels in response to the need for affordable and easily accessible medical tools. This invention effectively removes the need for intrusive blood draws and is particularly helpful for ongoing patient monitoring during phototherapy. Their hardware selections were influenced by the study's confirmation that bilirubin absorbs blue light at about 460

nm. They chose a stable LED design using RGB lights after experimenting with various configurations, observing that it was more processing-efficient than white light with diffraction gratings. Finally, they showed that there is a detectable change in the intensity of reflected light, especially in the blue and green spectrum, as bilirubin concentration rises (between 1 and 20 mg/dL).

Reference [4] is about figuring out neonatal jaundice in newborns without having to do those invasive tests that can be rough. The authors came up with this hybrid AI setup, I think it uses deep transfer learning on images that combine the baby's skin and the eyes or something like that. It seems like merging those two views helps get a better picture overall. Reference [5] is about figuring out neonatal jaundice in newborns without having to do those invasive tests that can be rough. The authors came up with this hybrid AI setup, I think it uses deep transfer learning on images that combine the baby's skin and the eyes or something like that. It seems like merging those two views helps get a better picture overall. They tested it against some regular machine learning things, like Multi Layer Perceptron, Decision Tree, Random Forest, and XGBoost. The dataset had 1430 samples in total, split into 760 for training and 670 for testing. That part feels a bit uneven, but maybe it's just how they divided it.

When they compared everything, XGBoost ended up being the best one, with accuracy around 98.693%. Other models did okay, but not quite as high. I am not totally sure if that's the absolute top, but it stood out in the results anyway. Newborns often get this thing called neonatal jaundice, where their bilirubin levels go up too much. If nobody does anything about it, that can cause big problems like damage to the brain or even hearing issues. It is pretty common, I guess. The usual ways to check for it involve blood tests for bilirubin or using phototherapy lights. Those work okay, but they are kind of invasive, and they cost a lot, especially where resources are limited. In places without good medical setups, it is harder to get those done right away. This study looks at using AI to spot jaundice early without all that hassle. They talk about analyzing pictures from smartphones or other digital stuff. It seems like a non-invasive way that could be easier for everyone.

They went through more than 15 different studies on this. Deep learning models, like ResNet, and some transfer learning techniques come up a lot. Those get really good results in accuracy, plus high sensitivity and specificity. I think the main point is that AI could help catch it sooner and cut down on the load for healthcare. Especially in areas that do not have much access, it might make things more efficient. The findings point to that, yeah. [5] In one study they developed this computer vision system for handling neonatal jaundice without being invasive at all. [6] It basically looks at images to diagnose and then jumps right into treatment if needed. I think that's pretty cool how it ties those together. The way it works involves changing the images over to Lab color space first. Then it uses something called Otsu's thresholding on the B and G channels to spot if there's hyperbilirubinemia. Not totally sure

on all the details there but it seems to pick up the bilirubin levels okay. [7]

Once it detects something off, the hardware kicks in with an Arduino Uno. That powers up these blue LEDs for therapy and also sends out an SMS alert through GSM. Its straightforward like that. They tested it on 20 infants plus a bunch of different test images. It worked well for TSB levels at 14 mg/dL or higher, which is the key range. Proved effective, yeah. What stands out are the advantages, like how affordable it is compared to other options. No need for those color calibration cards either, which saves time. And detection happens in just one second, super fast. Some parts might get messy in real use though. Reference [7] shows the promise of using image processing and artificial intelligence together with physiological traits to create effective, non-invasive diagnostic protocols. Study [8] sharpens our understanding of risk factors by separating weight categories. Data from this group revealed that low birth weight is a major predictor of severe jaundice, while macrosomia did not show any links. The authors claim that these results require standardized screening guidelines to allow for timely intervention. When considered with earlier research, there is strong agreement that tracking birth parameters is essential for lowering neonatal risks.

A study at [9] Hayatabad Medical Complex in Peshawar looked at the incidence of hyperbilirubinemia in polycythemic newborns. The researchers focused on infants under one month old who had normal gestational maturity and weight. They excluded babies with underlying syndromes or blood disorders. The results showed that about 50% of polycythemic infants had jaundice, and roughly one-quarter of them developed kernicterus. The analysis found strong links between jaundice and factors like being male, experiencing birth asphyxia, having a cephalohematoma, and having high hemoglobin levels. The authors highlighted the need for early screening and intervention in polycythemic newborns to prevent serious neurological problems.

Study [10] aimed to find out how often jaundice occurs in gallstone-induced pancreatitis and how quickly it goes away. Researchers followed 52 patients in a surgical department and found jaundice in over 60% of cases. This was closely linked to distended gallbladders and elevated enzyme levels. The condition usually resolved fairly quickly, with an average duration of 4.25 days. Females aged 26 to 50 were the most affected group. Using a large dataset from the Korean Health Insurance Review and Assessment Service, [11] performed a retrospective cohort analysis on long-term issues related to neonatal jaundice. Participants were divided based on severity, particularly whether they needed treatment. The results showed that infants who required treatment had a much higher chance of developing asthma and allergic rhinitis during early childhood compared to those with mild or no jaundice. The authors suggest that severe hyperbilirubinemia may predict future allergic issues, which means these infants may need closer long-term monitoring.

A review by [12] closely looked at the shift from invasive serum sampling to non-invasive biosensing for detecting

bilirubin. While recognizing that traditional methods are accurate, the authors argue they do not allow for continuous monitoring without trauma. The paper lists new technologies, including optical, electrochemical, and piezoelectric systems, which show promise for real-time diagnosis. However, many of these devices are still in the pre-clinical stage because of issues like cost, biological interference, and the need for regulatory approval. Studying computer vision as an alternative to invasive diagnostics, [13] created a deep-learning framework to identify severe neonatal jaundice. The study evaluated four architectures: MobileNet, EfficientNet, Vision Transformer, and a custom CNN, after thorough preprocessing and data augmentation. While Vision Transformer and EfficientNet showed good performance (AUC 0.9), the custom CNN reached the highest diagnostic performance with an AUC of 0.93. Notably, the model maintained reliability across various skin tones and lighting conditions, suggesting it could be a strong alternative to serum bilirubin tests.

Although non-invasive techniques are becoming more popular, a literature review [14] shows that there are still challenges in clinical use. After looking at methods ranging from blood tests to machine-learning algorithms, the authors found that while non-invasive approaches are dependable, they face technical and regulatory issues. They specifically noted the need for better handling of skin-tone differences and larger, more diverse datasets. This review outlines the research gaps that need to be filled to achieve fully automated and accurate jaundice detection.

In [15], researchers studied ensemble learning by combining CNNs with XGBoost to detect jaundice from facial images. The study focused on preprocessing methods like contrast improvement and color-feature analysis (e.g., B/Y ratios) to boost input quality. This two-step model effectively reduced environmental influences such as lighting, indicating that hybrid models can provide a practical solution for non-invasive clinical diagnostics.

III. PROBLEM STATEMENT

This study is about creating a system that uses intelligence to find out if a baby has neonatal jaundice just by looking at the baby's face. The system needs to be simple to use. Not hurt the baby. It also needs to be affordable and work well. This is important because the current ways to check for jaundice are not very good. Some tests are painful. Some people are not very good at checking, by just looking at the baby's face. The new system will use intelligence and look at pictures of the baby's face to see if the baby has neonatal jaundice.

IV. METHODOLOGY

The way we made the jaundice detection application is simple. It has four parts: we get the data ready we make a model we teach the model and we put it on the web. The jaundice detection application is made up of these four parts: getting the data ready making the model training the model and putting the jaundice detection application on the web. The dataset includes baby facial images classified into normal, mild

and severe jaundice levels. Each image is converted into RGB format, resizing it to 224*224 pixel, and normal is in pixel values to the [0,1] range. Data enhancement techniques such as rotation, cropping and lighting adjustment are adopted to increase diversity and control model complexity.

For model development, a hybrid architecture combining a Vision Transformer (ViT) and a Multi-Layer Perceptron (MLP) is used. The ViT is used to divide the image into smaller patches and deploying multi-head self-attention to capture global relationships. The output of ViT which is features of images given to MLP classifier consists of dense layers mapping functions and dropout for improvement of model's generalization. While training, the model is improved using the Adam optimizer and logarithmic loss for classification. Training stability and generalization were improved using early stopping and dynamic learning rate adjustments. The final model was preserved in the keras format for deployment. As a final step, the trained model is interfaced with Flask web application. This application allows users to upload infants images, processes them, runs prediction and displays the resulting classification and score.

Formulas: 1) Image Patch Embedding (Vision Transformer)

If the input image has dimensions $H \times W \times C$, the number of patches is:

$$N = H \times W / P^2$$

where:

H, W = image height and width

P = patch size

N = total number of patches

2) MLP Classification Layer

The hidden layer output is

$$h = \sigma(Wx + b)$$

where:

W = weight matrix

x = input feature vector

b = bias

σ = ReLU activation

3) Accuracy

$$\text{Accuracy} = (TP + TN / TP + TN + FP + FN) \times 100\%$$

V. PROPOSED WORK

This project is developed to provide a practical and harmless method for detection of jaundice in infants by using artificial intelligence and mobile technology. Its main aim is to analyze infants' facial and eye images using mobile cameras and deep learning model to completely remove dependency on blood tests. The process starts with capturing the image through an user friendly web interface created with Flask. Infant's image is uploaded, which is analyzed by the trained Keras model. This model integrates a Vision Transformer (ViT) to extract rich visual features with a Multilayer Perceptron (MLP) for classification, allowing accurate analysis of the jaundice as mild, normal, or severe.

This approach helps in multiple ways: It is safe, affordable, and easy to access. JaundiceNet can be used without any

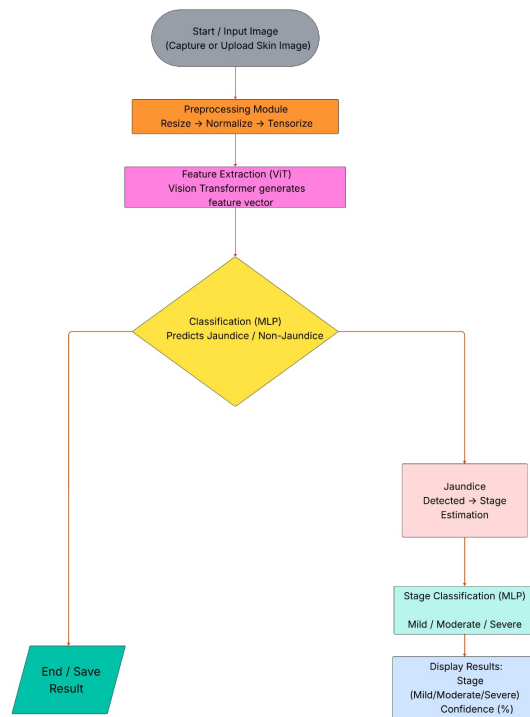


Fig. 1. Flow chart

special equipment or supervision. This system is scalable. It can be connected to cloud services. The cloud services help with monitoring and they provide security for the JaundiceNet data. The people who made JaundiceNet want to make it better in the future. They want to add pictures to the JaundiceNet system so it can work well with different skin colours and lighting conditions. They also want to add features like real-time warnings and support, for languages to the JaundiceNet system. The main goal of JaundiceNet is to work with systems that take care of newborn babies. JaundiceNet wants to help find jaundice and reduce the risks that come with it. JaundiceNet uses intelligence and mobile technology to make it easier to check babies for jaundice. This helps doctors and nurses give the babies the care they need when they need it. JaundiceNet wants to make it easier for people to get help when they need it.

VI. IMPLEMENTATION

Algorithm for JaundiceNet– ViT + MLP-Based Neonatal Jaundice Detection System

1. Start
2. Load Dependencies
Import required Python libraries
Flask (web framework)
PIL / OpenCV (image handling)
NumPy
PyTorch / TensorFlow (to load ViT + MLP model)
Load the pre-trained ViT–MLP hybrid model.
3. Initialize Flask Web Server
Create a Flask app instance.



Fig. 2. Infant Image

Configure file upload settings (allowed file types, upload directory).

4. Accept Image Input

User opens the web app.

User uploads an infant image (face/sclera).

Server receives the image and stores it temporarily.

5. Preprocess the Image

Load the uploaded image.

Convert the image to RGB format.

Resize the image to the model's required size (e.g., 224×224).

Normalize pixel values to [0, 1].

Convert the image into a tensor/batch format for the model.

6. Feature Extraction Using Vision Transformer (ViT)

Divide the image into small patches.

Embed patches and pass them through multi-head self attention layers, transformer encoder blocks, capturing local + global visual features.

Generate a feature vector representing the infant's eye/skin characteristics.

7. Classification Using MLP

Pass the ViT feature vector to the MLP.

Compute probability scores for classes:

Mild Jaundice

Normal / No Jaundice

Severe Jaundice

Use Softmax to obtain probability distribution.

Identify the class with the highest probability as final prediction.

8. Apply Thresholding (If Needed)

If the model outputs a single numerical value:

Apply thresholds (example):

value ; 0.3 → Normal

0.3 value ; 0.6 → Mild

value 0.6 → Severe

Convert numeric output into categorical prediction.

9. Generate Visualization Output

Display prediction to user:

Green → Normal

Yellow → Mild

Red → Severe

Show confidence percentage.

Return text + color-coded UI on HTML page.

10. Support Multi-Image Upload (Optional)

If user uploads multiple images:

Preprocess classify each image separately.

Store results in an array.

Display batch predictions.

11. Enable Scalability

Store uploaded images temporarily or push them to cloud storage.

Provide API endpoints for mobile app integration.

12.End

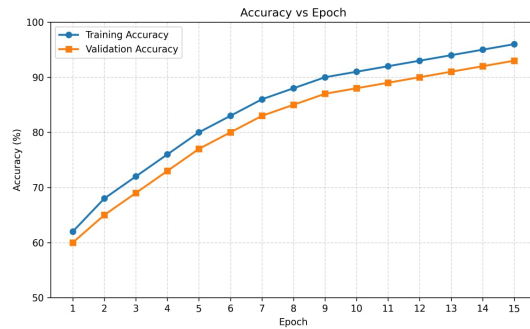


Fig. 3. Accuracy vs. Epoch

VII. PROPOSED WORK

VIII. RESULTS

The Jaundice detection model was successfully developed as an simple and user-friendly web application powered by ViT and MLP model. When user uploads image of the baby, the system automatically analyses the image according to the training and then predicts the jaundice level with a confidence score. The website displays the predicted category, image, and confidence level. In one of the test runs, the system predicted the category as Severe, with a confidence of 70.74. The results show that the integration of ViT for image features extraction with MLP classifier produces reliable and trustworthy predictions. The user-friendly interface, confidence levels, and explanations make this project useful for early detection and decision making, especially in regions with low medical facilities.

IX. CONCLUSION

The development of JaundiceNet marks an huge step towards accessible, harmless neonatal healthcare. By integrating deep learning with mobile technology, the system gives an practical solution by early jaundice detection, which removes the need for expensive blood tests and equipments. The in

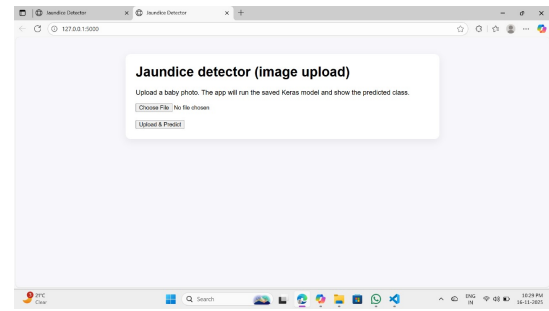


Fig. 4. Image Uploading

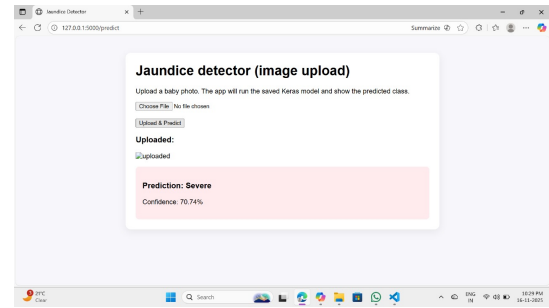


Fig. 5. Final Result

tegration of a Vision Transformer and Multilayer Perceptron enables an accurate classification of the severity of jaundice Fig. 4. Final Result. from simple facial or scleral images, making it suitable for use in both clinical and home environments. The user-friendly web application allows parents and healthcare providers to upload the image of the infant and get severity level predictions with confidence level, which enhances decision-making and timely medical interventions. This setup seems really helpful for rural spots, you know, where getting to a doctor takes forever or there are not many clinics around.

I think JaundiceNet is aiming bigger than just one thing. Its about using AI to check on baby health in general, not limited to jaundice stuff. Users get this tool thats straightforward to handle, doesnt cost a ton, and could grow with more features down the line. It feels like it puts some power back in peoples hands, especially if theyre not near big hospitals. Expanding the data collection is probably the main next step as it develops. Then theres adding cloud services, which might help with watching things from afar and keeping info safe. That part is kind of exciting, but im not sure how quickly it all comes together. Ultimately, It contributes so safer and easier neonatal healthcare by helping in early diagnosis, easy process, and linking the gaps in healthcare facilities—bringing advanced diagnosis closer to those who need them most.

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X. FUTURE WORK

Looking ahead, we are expanding the project capabilities to make it more personal and accessible. One of the key features is to get infant image on output screen. Another one of the most important and clear features is to download the report in PDF form which consists of infant's name, gender, age, medical record and parental details such as names of mother and father which helps to maintain each and every infant's data. By consolidating this information into a single, easy-to-share document, we aim to support continuity of care across different healthcare providers and empower families with clear, organized records of their child's health journey.

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